Records, Vitals and Band — QA Test Cases

Task 5.10 · v0.3 · Owner: Syed Sarim Jawad Zaidi · Sections A, A2 and B from docs/api/records_contract.md and appointment_contract.md (Tasks 5.1 and 5.2, 20 Oct 2026); C–D from SRS FR-H1, FR-H2 · 5.9 mockups v0.2

Purpose: prove that records can be added, viewed and exported by the right people only, that appointments remind both sides, that vitals are recorded and shown correctly, and that the band adds data without ever blocking the app. Results go on a copy of the QA Checklist template.

Status: Sections A (vault), A2 (record categories per Secondary) and B (appointments) use the real limits and error codes of Tasks 5.1 and 5.2. Sections C (vitals) and D (band) still have [bracketed] placeholders until Tasks 5.3 and 5.4 publish their contracts; v1.0 follows then.

1. Test setup

1.1 Test accounts

Role	Name in tests	Set-up
Dependent	Ammi	Registered, KYC verified, linked to Ali.
Primary Guardian	Ali	Phone + CNIC verified.
Secondary A	Sara	Invited by Ali, verified. Permission Health Records = ON.
Secondary B	Bilal	Invited by Ali, verified. Health Records OFF.
Outsider	Zain	A verified Guardian NOT linked to Ammi (for refusal checks).

1.2 How to run

- **Order:** backend checks through Postman as soon as Abdullah's 5.1–5.4 are merged; app checks after Talha's 5.6–5.8.
- **Test files:** one small PDF lab report, one phone photo of a prescription, one X-ray image, one non-medical file (e.g. .exe or .zip) and one very large file.
- **Band:** use mocked band data first (roadmap 5.4); repeat section D with the real band after 5.5.

2. Test cases

A. Medical vault (FR-G1, FR-G2, FR-G3)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
R-01	FR-G1	Ali logged in.	Medical Vault → Add record → Lab report, title "HbA1c", date, upload a PDF → Save.	201; record in the list (15) with category, record date and "added by Ali"; filter Lab shows it; file icon shown.		
R-02	FR-G1	Ammi logged in.	My health → My records → add a prescription photo with the camera.	Saved with "added by Ammi"; Ali sees it too.		
R-03	FR-G1	—	Add a record with only a note (no file); then one with neither file nor note.	Note-only record saved (no file icon; opening its file gives 404 NO_FILE). Empty one: 422 RECORD_NEEDS_CONTENT.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
R-04	FR-A4	Sara has health_records ON, Bilal OFF.	Both open Ammi's vault; then Ali switches Sara OFF and she refreshes.	Sara sees the records; Bilal gets 403 MODULE_ACCESS_DENIED and no Vault entry. After the switch-off Sara is refused on her next request.		
R-05	FR-G1	—	Upload a .exe renamed .pdf; a 12 MB photo; a 9 MB photo.	415 UNSUPPORTED_FILE (checked by the file's first bytes); 413 FILE_TOO_LARGE; the 9 MB photo is saved.		
R-06	FR-G1	Records added by Ali and by Sara.	Sara opens Ali's record and Ammi's record; then Ali opens Sara's record.	Sara: no Edit/Delete on others' records (can_modify false; a PATCH gives 403 RECORD_MODIFY_DENIED). Ali (Primary) can edit and delete any record.		
R-07	FR-G1	Phone network drops during an upload.	Retry the same upload (same client_id); then reuse that client_id for a different record.	Retry: 200 already_saved, only one record. Different record: 409 CLIENT_ID_REUSED.		
R-08	FR-G1	At least 6 records over 2 years, some with pictures and PDFs.	Export → Complete history → Make PDF.	Front page with name, period and a table of every record; pictures under their record; attached PDFs after the summary; a damaged PDF is listed as "Not included".		
R-09	FR-G1	Same.	Export One year (2026) and One month (Oct 2026); a month with no records; month without a year (Postman).	Only that period included; empty period still gives a clear PDF or message (record what happens); month without year: 422.		
R-10	FR-G1	A record with an Urdu title.	Export it.	Known limit: Urdu shows as ? inside the PDF (the app shows it correctly). Record as Low, not a failure.		
R-11	FR-G3	—	Make a doctor link (15f); open it in a private browser with no login; try to change anything.	201 with the link shown once; PDF opens read-only; Share / Copy work; the app adds the server address if the link starts with /.		
R-12	FR-G3	R-11 link.	Cancel the link and open it; make one with hours = 1 and open it after an hour; make 11 open links.	Cancelled: 410 SHARE_REVOKED; expired: 410 SHARE_EXPIRED; 11th link: 409 SHARE_LIMIT. Links list shows Active / Expired / Cancelled and open counts.		
R-13	FR-G3, A4	Sara made a link.	Ali switches Sara's health_records OFF; open Sara's link.	The link stops working.		
R-14	FR-N3	R-01 – R-12 done.	Ali opens the audit log (screen comes with Task 6.6; until then ask Abdullah to show the rows).	Every file read, export, add, change and delete is logged with who, what and when.		
R-15	FR-N2	—	Ask Abdullah to open a stored file directly on the server disk.	Unreadable (AES-256-GCM encrypted); the app still gets the original file.		
R-16	FR-G2	Task 5.2 built; details in A2.	Ali hides Prescriptions from Sara (15e); Sara opens the list, an export and a prescription by its id.	Prescriptions vanish from her list and exports; opening one by id gives 404 RECORD_NOT_FOUND (it looks like it does not exist).		

A2. Record categories per Secondary (FR-G2, Task 5.2)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
RA-01	FR-G2	Sara health_records ON; nothing hidden yet.	Sara opens Ammi's vault; GET …/records/access/me.	Sara sees all four categories (tabs and the category picker); access/me shows all four true. Nothing is hidden by default.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
RA-02	FR-G2	Prescriptions and lab reports exist.	Ali → Record access (15e) → Sara → Prescriptions OFF. Sara refreshes the list, filters by Prescriptions, exports a PDF, opens a prescription and its file by id (Postman), tries edit / delete.	PUT 200; on her next request prescriptions are gone from her list, total and export; the filter gives an empty list (not an error); open / file / edit / delete give 404 RECORD_NOT_FOUND. The Prescriptions tab and picker choice are not shown at all.		
RA-03	FR-G2	RA-02 done.	Sara adds a record as Prescription; moves a lab report into Prescription; makes a doctor link with categories = prescription.	Each gives 403 CATEGORY_NOT_ALLOWED; nothing is saved.		
RA-04	FR-G2, G3	Before RA-02, Sara made two doctor links: one for all categories, one for prescriptions only.	After RA-02, open both links.	The all-categories link opens without the prescriptions; the prescriptions-only link gives 410 SHARE_REVOKED.		
RA-05	FR-G2	Sara added a prescription herself before RA-02.	Sara looks for it.	Hidden from her too (agreed rule, clarification 105). Record as expected behaviour.		
RA-06	FR-G2	RA-02 done; switch Bilal's health_records ON for this case.	Bilal, Ali and Ammi open the vault.	Bilal still sees prescriptions (the setting is per Secondary); Ali (Primary) and Ammi (Dependent) always see all four.		
RA-07	FR-G2, A4	Bilal health_records OFF.	Ali opens Record access (15e).	Bilal's four switches are greyed out with "Needs Health Records permission first" (health_records_granted false).		
RA-08	FR-G2	—	Postman: Sara calls GET …/records/access and a PUT; Ali sends a PUT for himself; a PUT with a guardian_id that is not a Secondary; a PUT for category "photos"; the same PUT twice.	403 NOT_PRIMARY_GUARDIAN; 400 PRIMARY_NOT_GOVERNABLE; 404 SECONDARY_NOT_FOUND; 422 UNKNOWN_CATEGORY; repeating the same PUT gives the same answer.		
RA-09	FR-G2, N3	RA-02 done.	Ali turns Prescriptions back ON for Sara; check the audit rows.	Prescriptions come back on Sara's next request; audit shows record_category_hidden and record_category_shown with who and when.		

B. Appointments (FR-G4)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
AP-01	FR-G4	Ali logged in.	Appointments → Add: Dr. Hamid, tomorrow at this time + 5 min.	201; shows under Upcoming (15d) and on Ammi's My health (28); Ammi gets an appointment_update notification, Ali (who added it) does not.		
AP-02	FR-G4	AP-01 saved.	Wait for the reminders (or ask Abdullah to set a test appointment ~1 h ahead).	Ammi, Ali and Sara (health_records ON) get appointment_reminder 24 h and 1 h before; Bilal does not.		
AP-03	FR-G4	—	Add one in the past; add the same doctor at the same time twice; send a time without a time zone (Postman).	422 APPOINTMENT_IN_PAST; 409 APPOINTMENT_EXISTS; 422 VALIDATION_ERROR.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
AP-04	FR-G4	AP-01 saved.	Move it to another time; then cancel it; then try to change it.	Move: others notified, reminders follow the new time. Cancel: moves to History, others notified. Change after cancel: 409 APPOINTMENT_CANCELLED.		
AP-05	FR-G4	A cancelled and a past appointment.	Open History; delete one.	History shows both, latest first; delete removes it completely (204).		

C. Vitals (FR-H1)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
H-01	FR-H1	Ammi logged in.	My health → Add my blood pressure → 138 / 88 → Save.	Saved as manual, by Ammi, with time; Ali's Vitals (26) shows 138/88.		
H-02	FR-H1	Ali logged in.	Add reading (26b) → Blood sugar 112, SpO2 97, Temperature 98.4.	All saved, attributed to Ali; tiles update.		
H-03	FR-H1	—	Enter BP 400/20, SpO2 120 %, temperature 150, an empty value.	Each rejected with a clear message [valid ranges TBD].		
H-04	FR-H1	Readings on 7 different days.	Vitals → 7 days, then 30 days.	Chart points match the readings and dates; no missing or extra points.		
H-05	FR-H1	—	Look for any "blood pressure from band" option.	None: blood pressure is always typed in by hand.		

D. Wearable band (FR-H2, NFR-13)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
W-01	FR-H2	Band on and near Ammi's phone.	My band → Connect band (27b) → pick the band → Connect.	Paired; My band (27) shows Connected and last sync time.		
W-02	FR-H2	Paired.	Wait for a sync (or tap Sync now).	Heart rate and steps appear, marked Band; Guardian tile shows "Band · x min ago". Band battery is NOT in the app (3 LEDs on the band, team decision); check the LEDs by eye.		
W-03	FR-H2, NFR-13	Paired.	Switch the band off / walk out of range; use the app (Panic page, medicine, messages).	Everything else works; band status turns grey "Not connected · last sync …"; it reconnects by itself when back [time TBD].		
W-04	FR-H2	Paired.	Send a very high or very low heart rate from the mock data.	Shown, but no alarm or escalation starts from band data alone.		
W-05	Security	A second band nearby.	Try to pair Ammi's phone with a band that is not hers; check the app ignores data from an unexpected device.	Only the paired band's data is accepted (SDS: BLE bonding, no unexpected device address).		

3. Open points

- [Bracketed] values in sections C and D wait for Tasks 5.3 (vitals) and 5.4 (band sync): valid ranges, band reconnect time.

- Empty export period (R-09): the contract does not say what happens; record the behaviour and log it as a question if unclear.
- Band: custom ESP32 build by a local maker (sensor list to confirm); section D runs on mock data until the band arrives. Band battery is shown only by 3 LEDs on the band.
- Screen 15e lists each Secondary by name, but GET …/records/access (like the permissions endpoint) gives only email and phone; until a name is added the app shows the email.